

Viva Questions Topics

10/05/2026 20:58:43

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GIVE 20 ORAL questions on each of the following topic i will give you for...

Yes. Send me the topics one by one or all together. For each topic, I'll give...

Write a program non-recursive and recursive program to calculate...

Absolutely. Below are 20 viva/oral questions for EACH of the 6 practicals,...

1. Predict the price of the Uber ride from a given pickup point to the agree...

Absolutely. Here are 20 viva/oral questions for each of these 6 practicals,...

1. Installation of MetaMask and study spending Ether per transaction. 2....

Sure. Below are 20 viva/oral questions for EACH of these 6 Blockchain...

1. Write a program for Tracking Emails and Investigating Email Crimes. i....

Absolutely. Here are 20 viva/oral questions for each of these 7 practicals,...

1. Write a program to Compute Similarity between two text documents. 2...

Absolutely. Here are 20 viva/oral questions for each of these 6 Informatio...



GIVE 20 ORAL questions on each of the following topic i will give you for my practical exam

10/5/2026, 8:55:08 PM



Yes. Send me the **topics one by one or all together**.

For **each topic**, I'll give you:

- **20 oral/viva questions**
- **Short, easy-to-memorize answers**
- Questions ranging from **basic → practical → conceptual**
- Important questions likely to be asked in a **practical exam**
- Simple language suitable for viva preparation

Format:

Topic: Web Crawler

1. What is a web crawler?

→ A program that automatically visits web pages and collects information from them.

2. What is a seed URL?

→ The initial URL from which a web crawler starts crawling.

...up to **20 questions**.

Send me your **first topic**.

10/5/2026, 8:55:08 PM · gpt-5-6



Write a program non-recursive and recursive program to calculate Fibonacci numbers and analyze their time and space complexity. 2. Write a program to implement Huffman Encoding using a greedy strategy. 3. Write a program to solve a fractional Knapsack problem using a greedy method. 4. Write a program to solve a 0-1 Knapsack problem using dynamic programming or branch and bound strategy. 5. Design n-Queens matrix having first Queen placed. Use backtracking to place remaining Queens to generate the final n-queen's matrix. 6. Write a program for analysis of quick sort by using deterministic and randomized variant.

10/5/2026, 8:55:15 PM



Absolutely. Below are **20 viva/oral questions for EACH of the 6 practicals**, with **short, easy-to-remember answers**. I've focused on the questions an examiner is most likely to ask during a practical exam.

1. Fibonacci — Recursive & Non-Recursive

20 Viva Questions

1. What is the Fibonacci sequence?

→ A sequence where each number is the sum of the previous two numbers: 0, 1, 1, 2, 3, 5, 8...

2. What is the recursive formula for Fibonacci?

→ $F(n) = F(n-1) + F(n-2)$.

3. What are the base cases?

→ $F(0) = 0$ and $F(1) = 1$.

4. What is recursion?

→ A function calling itself to solve a smaller version of the problem.

5. **What is the time complexity of recursive Fibonacci?**
→ $O(2^n)$ approximately.
6. **Why is recursive Fibonacci inefficient?**
→ It calculates the same Fibonacci values repeatedly.
7. **What is the space complexity of recursive Fibonacci?**
→ $O(n)$ because of the recursion call stack.
8. **What is the time complexity of non-recursive Fibonacci?**
→ $O(n)$.
9. **What is the space complexity of iterative Fibonacci?**
→ $O(1)$ if only the previous two values are stored.
- L0. **Which Fibonacci implementation is faster?**
→ The non-recursive/iterative implementation.
- L1. **What is the difference between recursion and iteration?**
→ Recursion uses function calls; iteration uses loops.
- L2. **What is the first Fibonacci number?**
→ 0.
- L3. **What is F(5)?**
→ 5.
- L4. **What is F(10)?**
→ 55.
- L5. **Why does recursive Fibonacci have exponential complexity?**
→ Because each call generates two more recursive calls, causing repeated calculations.
- L6. **Can recursive Fibonacci be optimized?**
→ Yes, using **dynamic programming/memoization**.
- L7. **What is memoization?**
→ Storing already calculated results to avoid recalculating them.

18. **What is the best method for calculating large Fibonacci numbers?**
→ Iterative, memoization, or matrix-exponentiation methods are much better than simple recursion.
19. **What happens if n is very large in recursive Fibonacci?**
→ Execution becomes very slow and may cause stack overflow.
20. **What is the main purpose of this practical?**
→ To compare recursive and non-recursive approaches and analyze their time and space complexity.
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2. Huffman Encoding — Greedy Strategy

20 Viva Questions

1. **What is Huffman encoding?**
→ A lossless data compression technique that uses variable-length binary codes.
2. **What type of algorithm is Huffman coding?**
→ A **greedy algorithm**.
3. **Why is Huffman coding called greedy?**
→ It repeatedly selects the two symbols/nodes with the lowest frequencies.
4. **What is the main purpose of Huffman encoding?**
→ To reduce the number of bits required to represent data.
5. **What is a Huffman tree?**
→ A binary tree used to generate Huffman codes.
6. **What is stored in a leaf node?**
→ A character/symbol and its frequency.
7. **How do we select nodes during Huffman construction?**
→ We select the two nodes with the smallest frequencies.
8. **What do we assign to the left and right edges?**
→ Usually **0** to the left and **1** to the right.

9. **Is Huffman coding lossless?**
→ Yes.
10. **What is a prefix code?**
→ A code in which no code is the prefix of another code.
11. **Why are Huffman codes prefix-free?**
→ So that encoded data can be decoded without ambiguity.
12. **What data structure is commonly used to implement Huffman coding efficiently?**
→ A **min-priority queue/min-heap**.
13. **What is the time complexity of Huffman coding using a min-heap?**
→ **$O(n \log n)$** .
14. **What does frequency represent?**
→ The number of times a character occurs.
15. **Which characters get shorter codes?**
→ Characters with higher frequencies.
16. **Which characters get longer codes?**
→ Characters with lower frequencies.
17. **Is Huffman coding suitable for lossless compression?**
→ Yes.
18. **Can two characters have the same frequency?**
→ Yes.
19. **What is the root of a Huffman tree?**
→ The node containing the total frequency of all characters.
20. **Give one application of Huffman coding.**
→ It is used in compression formats such as **ZIP, JPEG and some multimedia compression systems**.

3. Fractional Knapsack — Greedy Method

20 Viva Questions

1. **What is the Fractional Knapsack problem?**
→ Selecting items to maximize profit when fractions of items are allowed.
2. **What is the main difference between fractional and 0-1 Knapsack?**
→ Fractional Knapsack allows items to be divided; 0-1 Knapsack does not.
3. **Which strategy is used for Fractional Knapsack?**
→ **Greedy strategy.**
4. **What is the greedy criterion?**
→ Select items according to the highest **profit/weight ratio**.
5. **What is profit/weight ratio?**
→ $\text{Profit} \div \text{Weight}$.
6. **Why do we calculate profit/weight ratio?**
→ To determine which item gives the maximum profit per unit weight.
7. **Can an item be partially selected?**
→ Yes.
8. **Can an item be completely selected?**
→ Yes.
9. **What happens when the remaining capacity is smaller than the item's weight?**
→ We take only the required fraction of that item.
10. **What is the time complexity?**
→ **$O(n \log n)$** because items are sorted by their ratio.
11. **What is the space complexity?**
→ Usually **$O(n)$** , depending on the implementation.
12. **Does greedy always give the optimal solution for Fractional Knapsack?**
→ Yes.

L3. Does greedy always work for 0-1 Knapsack?

→ No.

L4. What is the capacity of a knapsack?

→ The maximum weight it can hold.

L5. What is the objective of the problem?

→ Maximize total profit without exceeding capacity.

L6. What happens if an item's weight is exactly equal to remaining capacity?

→ The complete item is selected.

L7. What happens if the item has the highest ratio?

→ It is considered first.

L8. Why is sorting required?

→ To process items from highest to lowest profit/weight ratio.

L9. Is Fractional Knapsack an optimization problem?

→ Yes, it maximizes profit under a weight constraint.

L20. Why can't the same greedy approach be directly used for 0-1 Knapsack?

→ Because taking the highest ratio item may prevent a better combination of complete items.

4. 0-1 Knapsack — Dynamic Programming / Branch and Bound

20 Viva Questions

1. What is the 0-1 Knapsack problem?

→ Selecting complete items to maximize profit without exceeding capacity.

2. Why is it called 0-1 Knapsack?

→ Each item can either be selected (1) or not selected (0).

3. Can an item be divided?

→ No.

4. **What technique can be used to solve 0-1 Knapsack?**
→ Dynamic Programming or Branch and Bound.
5. **What is the DP recurrence for 0-1 Knapsack?**
→ $dp[i][w] = \max(dp[i-1][w], profit[i] + dp[i-1][w-weight[i]])$ when the item fits.
6. **What is the time complexity of DP Knapsack?**
→ $O(nW)$, where n is the number of items and w is capacity.
7. **What is the space complexity of the standard DP approach?**
→ $O(nW)$.
8. **What is the main difference between 0-1 and Fractional Knapsack?**
→ 0-1 doesn't allow fractions, while Fractional does.
9. **Does the greedy method always give the optimal 0-1 solution?**
→ No.
- L0. **What is the purpose of the DP table?**
→ To store the maximum profit for different numbers of items and capacities.
- L1. **What is overlapping subproblem?**
→ The same smaller problem is solved multiple times.
- L2. **What is optimal substructure?**
→ An optimal solution can be constructed from optimal solutions of smaller subproblems.
- L3. **Why is dynamic programming suitable here?**
→ Because the problem has overlapping subproblems and optimal substructure.
- L4. **What is Branch and Bound?**
→ A method that explores possible solutions while eliminating branches that cannot produce a better solution.
- L5. **What is a bound?**
→ An estimated maximum possible profit from a node.

L6. **What is pruning?**

→ Removing a branch that cannot produce a better solution.

L7. **What is the worst-case complexity of Branch and Bound?**

→ $O(2^n)$ in the worst case.

L8. **What is the advantage of DP over brute force?**

→ DP avoids repeatedly solving the same subproblems.

L9. **What happens if an item's weight exceeds the current capacity?**

→ That item cannot be included in that state.

20. **Which method is generally easier to implement for this practical?**

→ Dynamic Programming is usually simpler.

5. N-Queens — Backtracking

20 Viva Questions

1. **What is the N-Queens problem?**

→ Placing N queens on an $N \times N$ chessboard so that no two queens attack each other.

2. **What is the main technique used?**

→ **Backtracking.**

3. **How many queens are placed?**

→ N queens for an $N \times N$ board.

4. **Can two queens be in the same row?**

→ No.

5. **Can two queens be in the same column?**

→ No.

6. **Can two queens be on the same diagonal?**

→ No.

7. **What is backtracking?**
→ Trying a possible solution and undoing the choice when it leads to an invalid solution.
8. **Why is backtracking suitable for N-Queens?**
→ Because invalid arrangements can be rejected early.
9. **What is a safe position?**
→ A position where the queen is not attacked by any previously placed queen.
10. **How do we check the same column?**
→ Check whether another queen is already in that column.
11. **How do we check diagonals?**
→ By checking diagonal positions or using conditions based on row and column differences.
12. **What is the condition for two queens to be on the same diagonal?**
→ If $|row1 - row2| = |col1 - col2|$.
13. **What happens when no safe position is available?**
→ We backtrack to the previous queen and try another position.
14. **What is the base condition?**
→ When all N queens have been successfully placed.
15. **What is the time complexity of N-Queens backtracking?**
→ Approximately $O(N!)$ in the common analysis.
16. **What is the space complexity?**
→ $O(N^2)$ for the board, plus recursion/auxiliary space.
17. **For N=4, how many solutions exist?**
→ 2.
18. **For N=8, how many solutions exist?**
→ 92.

19. **Can N=2 be solved?**

→ No.

20. **Can N=3 be solved?**

→ No.

6. Quick Sort — Deterministic & Randomized

20 Viva Questions

1. **What is Quick Sort?**

→ A divide-and-conquer sorting algorithm based on partitioning an array around a pivot.

2. **What is a pivot?**

→ An element selected to divide the array into smaller and larger elements.

3. **What is partitioning?**

→ Rearranging elements so that elements smaller than the pivot are on one side and larger elements on the other.

4. **What is deterministic Quick Sort?**

→ Quick Sort where the pivot is selected using a fixed rule.

5. **Give examples of deterministic pivot selection.**

→ First element, last element, or middle element.

6. **What is randomized Quick Sort?**

→ Quick Sort where the pivot is selected randomly.

7. **Why use randomized Quick Sort?**

→ To reduce the chance of consistently getting bad partitions.

8. **What is the average time complexity of Quick Sort?**

→ $O(n \log n)$.

9. **What is the worst-case time complexity?**

→ $O(n^2)$.

L0. When does the worst case occur?

→ When partitions are highly unbalanced, such as sizes `0` and `n-1`.

L1. What is the best-case complexity?

→ $O(n \log n)$.

L2. What is the space complexity of Quick Sort?

→ Average recursion space is $O(\log n)$; worst case can be $O(n)$.

L3. Is Quick Sort a stable sorting algorithm?

→ Generally, no.

L4. Is Quick Sort an in-place sorting algorithm?

→ Yes, commonly it is considered in-place apart from recursion stack space.

L5. What is divide and conquer?

→ Divide the problem into smaller problems, solve them, and combine the results.

L6. What is the main difference between deterministic and randomized Quick Sort?

→ The pivot selection: fixed in deterministic, random in randomized.

L7. Can randomized Quick Sort still have $O(n^2)$ worst-case complexity?

→ Yes, theoretically, although the probability is very low for consistently bad choices.

L8. Why is Quick Sort generally fast in practice?

→ It has good average performance and good cache behavior with in-place partitioning.

L9. What happens after partitioning?

→ Quick Sort recursively sorts the left and right partitions.

20. Why is randomized Quick Sort useful for already sorted data?

→ Random pivot selection helps avoid consistently unbalanced partitions caused by fixed pivot rules.



Most Important Questions to Memorize



If your practical exam is short and you **cannot memorize all 120**, prioritize these:

Practical	Must-know concepts
Fibonacci	Recursive vs iterative, $O(2^n)$, $O(n)$, $O(1)$ space
Huffman	Greedy, min-heap, prefix code, $O(n \log n)$
Fractional Knapsack	Profit/weight ratio, greedy, fractions allowed, $O(n \log n)$
0-1 Knapsack	DP, recurrence, 0/1 choice, $O(nW)$, greedy doesn't always work
N-Queens	Backtracking, safe position, rows/columns/diagonals, $O(N!)$
Quick Sort	Pivot, partition, deterministic vs randomized, average $O(n \log n)$, worst $O(n^2)$

Especially memorize the time complexities—examiners very commonly ask them immediately after you run the program.

10/5/2026, 8:55:16 PM · gpt-5-6



1. Predict the price of the Uber ride from a given pickup point to the agreed drop-off location. Perform following tasks: 1. Pre-process the dataset. 2. Identify outliers. 3. Check the correlation. 4. Implement linear regression and random forest regression models. 5. Evaluate the models and compare their respective scores like R2, RMSE, etc. Dataset link: <https://www.kaggle.com/datasets/yasserh/uber-fares-dataset>

2. Classify the email using the binary classification method. Email Spam detection has two states: a) Normal State – Not Spam, b) Abnormal State – Spam. Use K-Nearest Neighbors and Support Vector Machine for classification. Analyze their performance. Dataset link: The emails.csv dataset on the Kaggle <https://www.kaggle.com/datasets/balaka18/email-spam-classification-dataset-csv>

3. Given a bank customer, build a neural network-based classifier that can determine whether they will leave or not in the next 6 months. Dataset Description: The case study is from an open-source dataset from Kaggle. The dataset contains 10,000 sample points with 14 distinct features such as CustomerId, CreditScore, Geography, Gender, Age, Tenure, Balance, etc. Link to the Kaggle project: <https://www.kaggle.com/barelydedicated/bank-customer-churn-modeling>

Perform following steps: 1. Read the dataset. 2. Distinguish the feature and target set and divide the data set into training and test sets. 3. Normalize the train and test data. 4. Initialize and build the model. Identify the points of improvement and implement the same. 5. Print the accuracy score and confusion matrix (5 points).

4. Implement Gradient Descent Algorithm to find the local minima of a function. For example, find the local minima of the function $y = (x+3)^2$ starting from the point $x=2$.

5. Implement K-Nearest Neighbors algorithm on diabetes.csv dataset. Compute confusion matrix, accuracy, error rate, precision and recall on the given dataset. Dataset link : <https://www.kaggle.com/datasets/abdallamahgoub/diabetes>

6. Implement K-Means clustering/ hierarchical clustering on sales_data_sample.csv dataset. Determine the

number of clusters using the elbow method. Dataset link : <https://www.kaggle.com/datasets/kyanyoga/sample-sales-data>

10/5/2026, 8:55:55 PM



Absolutely. Here are **20 viva/oral questions for each of these 6 practicals**, with **short, easy-to-memorize answers**. I've included the questions examiners are most likely to ask about **dataset preprocessing, algorithms, metrics, code, and outputs**.

1. Uber Fare Prediction — Linear Regression & Random Forest

20 Viva Questions

1. **What is the objective of this practical?**
→ To predict the Uber fare based on features such as pickup/drop-off location, passenger count, and distance-related information.
2. **What type of machine learning problem is this?**
→ **Supervised regression.**
3. **What is the target variable?**
→ The **fare amount**.
4. **What is data preprocessing?**
→ Cleaning and transforming raw data into a suitable format for machine learning.
5. **Why do we preprocess the dataset?**
→ To remove errors, missing values, outliers, and irrelevant data.
6. **What is an outlier?**
→ A data point that is significantly different from the other observations.
7. **How can outliers be detected?**
→ Using box plots, IQR, Z-score, or statistical methods.

8. **What is correlation?**

→ It measures the strength and direction of the relationship between two variables.

9. **What is the range of correlation coefficient?**

→ From **-1 to +1**.

L0. **What does correlation +1 mean?**

→ Perfect positive correlation.

L1. **What does correlation -1 mean?**

→ Perfect negative correlation.

L2. **What is Linear Regression?**

→ A supervised learning algorithm that predicts a continuous value using a linear relationship between variables.

L3. **What is Random Forest Regression?**

→ An ensemble regression method that combines predictions from multiple decision trees.

L4. **Why use Random Forest?**

→ It can model nonlinear relationships and is less sensitive to individual unusual observations than a single decision tree.

L5. **What is R^2 score?**

→ It measures how much of the variation in the target variable is explained by the model.

L6. **What is the ideal R^2 score?**

→ **1** is ideal; higher is generally better.

L7. **What is RMSE?**

→ Root Mean Squared Error, which measures the average magnitude of prediction errors.

L8. **Is lower or higher RMSE better?**

→ **Lower RMSE is better.**

19. **Which model can capture nonlinear relationships better: Linear Regression or Random Forest?**

→ Random Forest.

20. **Why do we compare both models?**

→ To determine which model predicts fares more accurately on the given dataset.

★ **Must remember**

Regression → continuous output

R^2 → higher is better

RMSE → lower is better

Correlation → -1 to +1

Random Forest → ensemble of decision trees

2. Email Spam Detection — KNN & SVM

20 Viva Questions

1. **What is the objective of this practical?**

→ To classify emails as **Spam or Not Spam**.

2. **What type of machine learning problem is this?**

→ **Binary classification**.

3. **What are the two classes?**

→ Spam and Not Spam.

4. **What is spam?**

→ An unwanted or potentially harmful email, often sent in bulk.

5. **What is binary classification?**

→ Classification involving only two possible classes.

6. **What is KNN?**

→ K-Nearest Neighbors is an algorithm that classifies a data point based on its nearest neighboring points.

7. **What does K represent in KNN?**
 - The number of nearest neighbors considered.
8. **How does KNN classify an email?**
 - It finds the K most similar training examples and assigns the class based on their votes.
9. **What happens if K=1?**
 - The class of the nearest training point is assigned.
10. **What is SVM?**
 - Support Vector Machine is a classification algorithm that finds a decision boundary separating classes.
11. **What is a hyperplane?**
 - A boundary that separates different classes in the feature space.
12. **What is a support vector?**
 - A data point close to the decision boundary that influences the position of the boundary.
13. **What is the margin in SVM?**
 - The distance between the decision boundary and the closest data points from each class.
14. **Why is feature scaling important for KNN?**
 - Because KNN uses distance, and features with larger numerical values can dominate the distance calculation.
15. **Is feature scaling useful for SVM?**
 - Yes, especially when features have different scales.
16. **What is accuracy?**
 - The proportion of correctly classified samples out of all samples.
17. **What is precision?**
 - The proportion of predicted positive samples that are actually positive.

18. **What is recall?**

→ The proportion of actual positive samples correctly identified.

19. **Which metric can be especially important in spam detection?**

→ Precision and recall can both be important, depending on whether false spam alerts or missed spam is more costly.

20. **Why compare KNN and SVM?**

→ To determine which classification algorithm performs better on the email dataset.

★ **Must remember**

KNN → nearest neighbors

SVM → separating hyperplane

K → number of neighbors

Support vectors → closest important points

Spam detection → binary classification

3. Bank Customer Churn — Neural Network

20 Viva Questions

1. **What is customer churn?**

→ When a customer leaves or stops using a company's service.

2. **What is the objective of this practical?**

→ To predict whether a bank customer will leave the bank.

3. **What type of machine learning problem is this?**

→ **Binary classification.**

4. **What is the target variable?**

→ The **Exited** variable.

5. **What does Exited = 1 mean?**

→ The customer left the bank.

6. **What does Exited = 0 mean?**

→ The customer did not leave.

7. **What is a neural network?**
→ A machine learning model made of interconnected artificial neurons arranged in layers.
8. **What are the main layers of a neural network?**
→ Input layer, hidden layer(s), and output layer.
9. **What is an input feature?**
→ A variable used by the model to make a prediction.
10. **Why do we divide the dataset into training and testing data?**
→ Training data is used to learn the model, while test data is used to evaluate its performance on unseen data.
11. **What is normalization?**
→ Scaling numerical features to a common range or scale.
12. **Why is normalization important for neural networks?**
→ It helps training become more stable and efficient.
13. **What is an activation function?**
→ A function that introduces nonlinearity into a neural network.
14. **Name some activation functions.**
→ ReLU, sigmoid, and tanh.
15. **Which activation function is commonly used in the output layer for binary classification?**
→ **Sigmoid.**
16. **What does sigmoid produce?**
→ A value between **0 and 1**, which can be interpreted as a probability.
17. **What is an epoch?**
→ One complete pass through the training dataset.
18. **What is a confusion matrix?**
→ A table showing correct and incorrect predictions for each class.

19. **What is accuracy?**

→ The percentage of total predictions that are correct.

20. **How can a neural network be improved?**

→ By tuning layers, neurons, learning rate, epochs, batch size, activation functions, or using regularization/dropout.

★ **Must remember**

Churn → customer leaving

Exited → target

Sigmoid → binary output

Normalization → common scale

Confusion matrix → prediction summary

4. Gradient Descent — Local Minimum

Function:

$y = (x + 3)^2$, starting from $x = 2$

20 Viva Questions

1. **What is Gradient Descent?**

→ An optimization algorithm used to find a minimum of a function.

2. **What is the objective of this practical?**

→ To find the local minimum of a given function using Gradient Descent.

3. **What is the given function?**

→ $y = (x + 3)^2$.

4. **What is the derivative of the function?**

→ $dy/dx = 2(x + 3)$.

5. **What is the minimum point of this function?**

→ $x = -3$.

6. **What is the minimum value of y?**

→ 0.

7. What is a gradient?

→ The derivative that indicates the direction and rate of change of a function.

8. What is the Gradient Descent update equation?

→ $x_{\text{new}} = x_{\text{old}} - \text{learning_rate} \times \text{gradient}$.

9. What is the learning rate?

→ It controls how large each update step is.

L0. What happens if the learning rate is too large?

→ The algorithm may overshoot the minimum or fail to converge.

L1. What happens if the learning rate is too small?

→ Convergence becomes very slow.

L2. What is convergence?

→ The process of approaching a stable minimum value.

L3. What is a local minimum?

→ A point where the function value is lower than nearby points.

L4. What is the starting point in this problem?

→ $x = 2$.

L5. What happens to x during Gradient Descent?

→ It moves toward -3 .

L6. What is the gradient at $x = 2$?

→ $2(2+3) = 10$.

L7. What is the gradient at $x = -3$?

→ 0 .

L8. Why does Gradient Descent stop at $x = -3$?

→ Because the gradient becomes zero at the minimum.

L9. Where is Gradient Descent commonly used?

→ In machine learning model optimization, especially for minimizing loss functions.

20. Is Gradient Descent a machine learning model?

→ No. It is an **optimization algorithm** used to train/optimize models.

★ Must remember

Formula:

$x_{\text{new}} = x_{\text{old}} - \text{learning rate} \times \text{gradient}$

For $y = (x+3)^2$:

$\text{gradient} = 2(x+3)$

Minimum: $x = -3, y = 0$

5. K-Nearest Neighbors — Diabetes Dataset

20 Viva Questions

1. What is the objective of this practical?

→ To classify diabetes outcomes using the KNN algorithm.

2. What type of problem is this?

→ **Binary classification.**

3. What is KNN?

→ A classification algorithm that predicts a sample based on its nearest neighbors.

4. What does K represent?

→ Number of nearest neighbors considered.

5. How does KNN calculate similarity?

→ Usually using a distance measure such as Euclidean distance.

6. What is Euclidean distance?

→ The straight-line distance between two points.

7. Why is feature scaling important in KNN?

→ Because KNN is distance-based.

8. What happens with a very small K?

→ The model may become sensitive to noise.

9. **What happens with a very large K?**
→ The model may become too generalized and lose important local patterns.
10. **What is a confusion matrix?**
→ A table containing TP, TN, FP, and FN.
11. **What is True Positive?**
→ A positive case correctly predicted as positive.
12. **What is True Negative?**
→ A negative case correctly predicted as negative.
13. **What is False Positive?**
→ A negative case incorrectly predicted as positive.
14. **What is False Negative?**
→ A positive case incorrectly predicted as negative.
15. **What is accuracy?**
→ $(TP + TN) / (TP + TN + FP + FN)$.
16. **What is error rate?**
→ The proportion of incorrect predictions.
17. **What is precision?**
→ $TP / (TP + FP)$.
18. **What is recall?**
→ $TP / (TP + FN)$.
19. **What does high recall mean in diabetes detection?**
→ Most actual diabetic cases are correctly identified.
20. **What is one disadvantage of KNN?**
→ Prediction can be computationally expensive for large datasets because it compares a new point with training points.

★ **Must remember**

KNN → distance-based

Small K → sensitive to noise

Large K → smoother/generalized

Precision = $TP/(TP+FP)$

Recall = $TP/(TP+FN)$

Accuracy = $(TP+TN)/Total$

6. K-Means / Hierarchical Clustering — Sales Dataset

20 Viva Questions

1. **What is clustering?**

→ Grouping similar data points together without predefined class labels.

2. **What type of machine learning is clustering?**

→ **Unsupervised learning.**

3. **What is K-Means clustering?**

→ An algorithm that divides data into K clusters based on similarity.

4. **What does K represent?**

→ The number of clusters.

5. **What is a centroid?**

→ The center point of a cluster.

6. **How does K-Means work?**

→ Initialize centroids, assign points to the nearest centroid, update centroids, and repeat until convergence.

7. **What is the objective of K-Means?**

→ Minimize the distance between data points and their assigned cluster centroids.

8. **What is the elbow method?**

→ A technique used to determine a suitable number of clusters.

9. **Why is it called the elbow method?**
 - The plot of within-cluster error versus K often forms an elbow at a suitable K.
10. **What is inertia/WCSS?**
 - The sum of squared distances between data points and their cluster centroids.
11. **What does a lower WCSS mean?**
 - Data points are closer to their cluster centroids.
12. **Why can't we simply choose the largest K?**
 - Increasing K generally reduces within-cluster error but can create unnecessarily many clusters.
13. **What is hierarchical clustering?**
 - A clustering technique that builds a hierarchy of clusters.
14. **What is a dendrogram?**
 - A tree-like diagram showing how clusters are merged or divided.
15. **What is agglomerative clustering?**
 - A bottom-up approach where each point starts as a separate cluster and clusters are repeatedly merged.
16. **What is divisive clustering?**
 - A top-down approach where one cluster is repeatedly divided.
17. **What is the difference between K-Means and hierarchical clustering?**
 - K-Means requires the number of clusters beforehand, while hierarchical clustering can visualize cluster relationships using a dendrogram.
18. **Why do we scale data before clustering?**
 - To prevent large-scale features from dominating distance calculations.
19. **Is K-Means supervised or unsupervised?**
 - **Unsupervised.**
20. **What is the purpose of clustering sales data?**
 - To identify groups of similar sales patterns, customers, products, or markets.

★ Must remember

Clustering → unsupervised

K-Means → centroid-based

K → number of clusters

Elbow method → choose K

WCSS/Inertia → within-cluster error

Hierarchical → dendrogram

🔥 MOST IMPORTANT VIVA QUESTIONS FROM ALL 6

If your examiner gives you very little time, **memorize these first:**

Topic	Question	Answer
Uber	What type of problem?	Regression
Uber	Best R^2 ?	1
Uber	RMSE better higher/lower?	Lower
Uber	Random Forest?	Ensemble of decision trees
Spam	What type?	Binary classification
Spam	KNN?	Nearest-neighbor classification
Spam	SVM?	Finds a separating hyperplane
Churn	Target?	Exited
Churn	Problem type?	Binary classification
Churn	Output activation?	Sigmoid
Churn	Why normalize?	Put features on comparable scales
Gradient Descent	Purpose?	Find/minimize a function
Gradient Descent	Update formula?	$x = x - \text{learning rate} \times \text{gradient}$
Gradient Descent	Minimum of $(x+3)^2$?	$x = -3$

Topic	Question	Answer
Diabetes	KNN based on?	Distance
Diabetes	Precision?	$TP/(TP+FP)$
Diabetes	Recall?	$TP/(TP+FN)$
Clustering	Type?	Unsupervised
Clustering	K-Means uses?	Centroids
Clustering	Elbow method?	Finds suitable number of clusters

One very important tip for your practical exam: for each program, be ready to explain **(1) dataset, (2) preprocessing, (3) algorithm, (4) why that algorithm is used, (5) important parameters, (6) output/metrics, and (7) time/space or advantages/disadvantages**. Those 7 areas cover a large portion of typical practical viva questions.

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1. Installation of MetaMask and study spending Ether per transaction. 2. Create your own wallet using Metamask for crypto transactions. 3. Write a smart contract on a test network, for Bank account of a customer for following operations: ☐ Deposit money ☐ Withdraw Money ☐ Show balance 4. Write a program in solidity to create Student data. Use the following constructs: ☐ Structures ☐ Arrays ☐ Fallback Deploy this as smart contract on Ethereum and Observe the transaction fee and Gas values. 5. Write a survey report on types of Blockchains and its real time use cases. 6. Write a program to create a Business Network using Hyperledger

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Sure. Below are **20 viva/oral questions for EACH of these 6 Blockchain practicals**, with **short, simple answers** that are easy to memorize for your practical exam.

1. MetaMask Installation & Study of Ether Spent per Transaction

20 Viva Questions

1. What is MetaMask?

→ MetaMask is a cryptocurrency wallet that allows users to interact with blockchain networks such as Ethereum.

2. What is a crypto wallet?

→ A wallet is an application that manages blockchain accounts, keys, and transactions.

3. What is Ether?

→ Ether (ETH) is the native cryptocurrency of the Ethereum network.

4. **What is Ethereum?**

→ Ethereum is a blockchain platform that supports cryptocurrency and smart contracts.

5. **What is a transaction?**

→ A transaction is an operation recorded on a blockchain, such as transferring ETH.

6. **What is a transaction fee?**

→ It is the fee paid to the blockchain network for processing a transaction.

7. **What is gas?**

→ Gas measures the computational work required to execute an Ethereum transaction or smart contract operation.

8. **What is gas price?**

→ Gas price is the amount paid per unit of gas.

9. **What is the basic relationship between gas and transaction fee?**

→ Transaction fee depends on the gas used and the gas price.

10. **Who receives the transaction fee?**

→ Validators receive transaction-related rewards/fees according to Ethereum's current fee mechanism.

11. **What is a block?**

→ A block is a group of blockchain transactions and related data.

12. **What is a blockchain explorer?**

→ A website/tool used to view transactions, blocks, addresses, and other blockchain information.

13. **What is a transaction hash?**

→ A unique identifier for a blockchain transaction.

14. **Can a confirmed Ethereum transaction normally be reversed?**

→ No. Blockchain transactions are generally irreversible once confirmed.

15. **What is a public address?**

→ An address that can be shared with others to receive cryptocurrency.

L6. What is a private key?

→ A secret cryptographic key used to authorize transactions.

L7. Should a private key or seed phrase be shared?

→ **Never.**

L8. What is a seed phrase?

→ A sequence of words used to recover a wallet.

L9. Why should a test network be used for practical experiments?

→ It allows experimentation without risking real funds.

20. What do you observe when studying an Ether transaction?

→ Transaction hash, sender, receiver, amount, gas used, gas price, and transaction fee.

2. Create Your Own Wallet Using MetaMask

20 Viva Questions

1. What is a wallet in blockchain?

→ It is a tool for managing blockchain accounts and cryptographic keys.

2. What is MetaMask used for?

→ It is used to manage Ethereum-compatible accounts and interact with decentralized applications.

3. What is an Ethereum address?

→ It is a public identifier used to send and receive assets on Ethereum.

4. Can the wallet address be shared?

→ Yes, the public address can be shared for receiving funds.

5. Can the private key be shared?

→ No.

6. What is a seed/recovery phrase?

→ A set of words used to restore a wallet.

7. **Why is the recovery phrase important?**
 - Anyone who possesses it may be able to control the associated wallet.
8. **What is a public key?**
 - A cryptographic key related to the wallet that can be used to derive/address identification.
9. **What is a private key?**
 - A secret key used to sign transactions.
10. **What is a digital signature?**
 - A cryptographic proof that a transaction was authorized by the account holder.
11. **What is an account address?**
 - A unique blockchain address associated with an account.
12. **What is ETH used for?**
 - It is used for value transfer and paying transaction fees on Ethereum.
13. **What is gas?**
 - A unit representing computational work on Ethereum.
14. **What is a testnet?**
 - A blockchain network intended for testing applications and transactions.
15. **Why should students use testnet ETH?**
 - It has no real monetary value and is useful for testing.
16. **What is a dApp?**
 - A decentralized application that interacts with blockchain smart contracts.
17. **How does MetaMask connect to a dApp?**
 - It allows the dApp to request blockchain interactions and transaction signatures from the user.
18. **What happens when you send ETH?**
 - MetaMask creates/signs a transaction and broadcasts it to the network.

19. **What is a transaction hash?**

→ A unique identifier used to track a transaction.

20. **What is the most important security rule for MetaMask?**

→ Never share the recovery phrase or private key.

3. Smart Contract — Bank Account

Operations:

- Deposit
- Withdraw
- Show Balance

20 Viva Questions

1. **What is a smart contract?**

→ A program deployed on a blockchain that executes according to predefined rules.

2. **Which language is commonly used to write Ethereum smart contracts?**

→ **Solidity.**

3. **What is the purpose of this smart contract?**

→ To simulate basic bank account operations using blockchain.

4. **What is the deposit function?**

→ It adds cryptocurrency to the contract/account balance.

5. **What is the withdrawal function?**

→ It transfers funds out of the contract after checking the required conditions.

6. **What does the balance function do?**

→ It returns the current balance.

7. **What is `msg.sender` ?**

→ The address that initiated the current call or transaction.

8. **What is `msg.value` ?**
→ The amount of native cryptocurrency sent with a payable transaction.
9. **What is `payable` in Solidity?**
→ A function or address that can receive native cryptocurrency.
10. **Why is the deposit function generally payable?**
→ Because it needs to receive ETH.
11. **What happens if a user tries to withdraw more than their balance?**
→ The transaction should fail/revert.
12. **What is `require()` used for?**
→ To check conditions and revert execution when a condition is false.
13. **What is a transaction?**
→ A blockchain operation that changes state and generally requires gas.
14. **What is a smart contract address?**
→ The blockchain address where the deployed contract resides.
15. **What is contract state?**
→ Data stored by the contract on the blockchain.
16. **What is gas in smart contracts?**
→ The computational resource consumed when executing contract operations.
17. **Why does deploying a contract require gas?**
→ Because deployment requires blockchain computation and storage.
18. **What is a test network?**
→ A network used to test contracts without using real-value funds.
19. **What happens after a smart contract is deployed?**
→ It receives an address and its code/state becomes available on the blockchain.
20. **What is one advantage of a blockchain-based bank contract?**
→ Transactions and contract state can be transparently recorded and verified on the blockchain.

4. Solidity Student Data — Structures, Arrays & Fallback

20 Viva Questions

- 1. What is a structure in Solidity?**
→ A user-defined data type that groups related variables.
- 2. Why do we use structures?**
→ To store multiple related pieces of information together.
- 3. Give an example of student data.**
→ Roll number, name, age, and marks.
- 4. What is an array?**
→ A collection of elements of the same type.
- 5. What is a dynamic array?**
→ An array whose size can change during contract execution.
- 6. What is a fixed-size array?**
→ An array with a predefined size.
- 7. Why are arrays used in the student contract?**
→ To store multiple student records.
- 8. What is a fallback function?**
→ A special function invoked when a contract receives a call that doesn't match an existing function, subject to Solidity's fallback rules.
- 9. When is a fallback function called?**
→ When no matching function exists for the call data, or when a plain Ether transfer occurs and no receive function is present.
- 10. What is the `receive()` function?**
→ A special function designed to receive plain Ether transfers when no calldata is supplied.

L1. What is the difference between fallback and receive?

→ `receive()` handles plain Ether transfers without calldata; `fallback()` handles unmatched calls and can also receive Ether if payable.

L2. What is Solidity?

→ A programming language used to write smart contracts for Ethereum-compatible blockchains.

L3. What is gas?

→ A measure of computational work used by Ethereum operations.

L4. Why does storing student data consume gas?

→ Because changing blockchain state requires computation and storage.

L5. What is a transaction fee?

→ The fee paid for processing a blockchain transaction.

L6. What is gas limit?

→ The maximum amount of gas a transaction is allowed to consume.

L7. What is gas used?

→ The actual amount of gas consumed during execution.

L8. What is a transaction hash?

→ A unique identifier of a blockchain transaction.

L9. Where can you observe gas values?

→ In a blockchain explorer or wallet transaction details.

20. Why is gas important in this practical?

→ It shows the computational/storage cost of deploying and interacting with the smart contract.

5. Survey — Types of Blockchains & Real-Time Use Cases

20 Viva Questions

1. **What is blockchain?**
→ A distributed digital ledger that records transactions in linked blocks.
2. **What are the main types of blockchain?**
→ Public, private, consortium, and hybrid blockchains.
3. **What is a public blockchain?**
→ A blockchain generally open for anyone to participate in according to its protocol.
4. **Give an example of a public blockchain.**
→ Ethereum.
5. **What is a private blockchain?**
→ A blockchain where participation is controlled by an organization.
6. **What is a consortium blockchain?**
→ A blockchain governed by a group of organizations.
7. **What is a hybrid blockchain?**
→ A combination of public and private blockchain characteristics.
8. **What is decentralization?**
→ Distributing control and decision-making across multiple participants rather than one central authority.
9. **What is immutability?**
→ The difficulty of altering confirmed blockchain records without detection or consensus.
10. **What is a distributed ledger?**
→ A ledger replicated and maintained across multiple network participants.
11. **What is consensus?**
→ The mechanism by which blockchain participants agree on the valid state of the ledger.
12. **Name two consensus mechanisms.**
→ Proof of Work and Proof of Stake.

L3. What is Proof of Work?

→ A consensus mechanism where participants prove computational work to help secure the network.

L4. What is Proof of Stake?

→ A consensus mechanism where validators participate based on staked cryptocurrency and protocol rules.

L5. What are smart contracts?

→ Programs stored and executed on a blockchain.

L6. Give a blockchain use case in supply chain.

→ Tracking products and transactions through different stages of the supply chain.

L7. How can blockchain be used in healthcare?

→ It can support controlled sharing and verification of health-related records.

L8. How can blockchain be used in banking?

→ For areas such as digital assets, settlement, and transaction recording.

L9. What is the main advantage of blockchain?

→ It can provide shared, tamper-evident records without relying on a single central database.

L10. What is one limitation of blockchain?

→ Scalability, transaction costs, privacy, and energy use can be challenges depending on the blockchain design.

6. Business Network Using Hyperledger

20 Viva Questions

1. What is Hyperledger?

→ An open-source umbrella project hosted by the Linux Foundation for enterprise blockchain technologies.

2. What is Hyperledger Fabric?

→ A permissioned blockchain platform designed for enterprise applications.

3. **What is a business network?**

→ A network in which organizations and participants interact to exchange assets or perform business transactions.

4. **Is Hyperledger Fabric public or permissioned?**

→ **Permissioned.**

5. **What does permissioned blockchain mean?**

→ Only authorized participants can access and perform permitted operations.

6. **What is a peer in Hyperledger Fabric?**

→ A network component that hosts ledgers and smart contracts/chaincode and participates in transactions.

7. **What is an organization?**

→ An entity participating in the Fabric network.

8. **What is a ledger?**

→ The record of blockchain transactions and current state.

9. **What is chaincode?**

→ The smart-contract code that defines business logic in Hyperledger Fabric.

10. **What is a channel?**

→ A private communication and ledger environment for a subset of network members.

11. **What is an ordering service?**

→ A component/service that orders transactions into blocks for the network.

12. **What is an MSP?**

→ Membership Service Provider; it manages identities and membership rules for network participants.

13. **Why does Hyperledger use identities?**

→ To authenticate and authorize known participants.

14. **What is a certificate authority?**

→ A component that issues digital certificates used to establish participant

identities.

L5. What is Docker commonly used for in Fabric?

→ Running network components in containers.

L6. What is the purpose of the ledger?

→ To maintain transaction history and current state.

L7. What is a transaction in Hyperledger Fabric?

→ A proposed operation that is endorsed, ordered, and committed according to the network's transaction flow.

L8. What is endorsement?

→ Approval of a transaction proposal by the required authorized peers according to the endorsement policy.

L9. Why is Hyperledger suitable for businesses?

→ It provides permissioned access, identity management, privacy features, and configurable governance.

20. Give one use case of Hyperledger Fabric.

→ Supply-chain tracking, trade finance, healthcare data sharing, or inter-organizational asset management.



TOP 30 QUESTIONS TO MEMORIZE FIRST

If you have **very little time before the practical**, memorize these:

MetaMask

1. **What is MetaMask?** → A crypto wallet and dApp interface.
2. **What is ETH?** → Native cryptocurrency of Ethereum.
3. **What is gas?** → Measure of computational work.
4. **What is transaction fee?** → Fee for processing a transaction.
5. **What is a seed phrase?** → Wallet recovery phrase; never share it.

Smart Contracts



6. **What is a smart contract?** → Program running on a blockchain.
7. **What language is used?** → Solidity.
8. **What is `msg.sender`?** → Address initiating the call/transaction.
9. **What is `msg.value`?** → ETH sent with a payable call.
10. **What is `require()`?** → Checks a condition and reverts if false.

Solidity

- L1. **What is a structure?** → Groups related variables.
- L2. **What is an array?** → Collection of same-type elements.
- L3. **What is fallback?** → Handles unmatched calls under Solidity's fallback rules.
- L4. **What is `receive()`?** → Handles plain ETH transfers without calldata.
- L5. **Why does contract execution need gas?** → It consumes blockchain computation/resources.

Blockchain

- L6. **Types of blockchain?** → Public, private, consortium, hybrid.
- L7. **What is public blockchain?** → Generally open participation.
- L8. **What is private blockchain?** → Restricted participation.
- L9. **What is consensus?** → Agreement on valid blockchain state.
20. **Name consensus mechanisms.** → PoW and PoS.

Hyperledger

21. **What is Hyperledger?** → Open-source enterprise blockchain project ecosystem.
22. **What is Hyperledger Fabric?** → Permissioned enterprise blockchain platform.
23. **What is chaincode?** → Smart-contract/business logic in Fabric.
24. **What is a peer?** → Component that maintains ledger and runs chaincode.
25. **What is a channel?** → Private blockchain/communication environment for selected members.

Very Common Examiner Traps

- 26. **Can you share your private key?** → Never.
- 27. **Can you share your seed phrase?** → Never.
- 28. **Does MetaMask itself store your cryptocurrency?** → It manages keys/accounts and interfaces with blockchain networks; assets are recorded on-chain.
- 29. **Is Hyperledger Fabric permissionless like a public blockchain?** → No, it is permissioned.
- 30. **Does higher gas always mean a better transaction?** → No. Gas relates to computational work/limits and gas price affects fees; it isn't a measure of transaction quality.

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1. Write a program for Tracking Emails and Investigating Email Crimes. i.e. Write a program to analyze e-mail header 2. Implement a program to generate and verify CAPTCHA image 3. A person on a nearby road is trying to enter into a WiFi network by trying to crack the Password to use the IP Printer resource; write a program detect such attempt and prohibit the access. Develop the necessary scenario by Using an IEEE 802.11, configure a Wi-Fi adapter and Access Point 4. Write a computer forensic application program for Recovering permanent Deleted Files and Deleted Partitions 5. Write a program for Log Capturing and Event Correlation 6. Configure and demonstrate use of vulnerability assessment tool like Wireshark or SNORT 7. Study of Honeypot

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Absolutely. Here are **20 viva/oral questions for each of these 7 practicals**, with **short and easy answers** for your practical exam.

1. Email Tracking & Investigation — Email Header Analysis

20 Viva Questions

1. **What is an email header?**

→ Metadata containing information about an email, such as sender, receiver, date, and routing details.

2. **Why is email header analysis important in forensics?**

→ It helps investigate the origin, route, and authenticity of an email.

3. **What is the `From` field?**

→ It shows the sender address displayed in the email.



4. **What is the To field?**
→ It shows the intended recipient.
5. **What is the Date field?**
→ It indicates when the email was sent according to the header.
6. **What is the Subject field?**
→ It contains the subject of the email.
7. **What is the Received header?**
→ It records information about servers that handled the email.
8. **Why are multiple Received headers present?**
→ Because an email can pass through multiple mail servers.
9. **What is an IP address in an email header?**
→ It identifies a network endpoint associated with a mail server or connection.
10. **What is SMTP?**
→ Simple Mail Transfer Protocol, commonly used to send email.
11. **What is an email spoofing attack?**
→ Forging email information to make a message appear to come from another sender.
12. **Can the From address alone prove the actual sender?**
→ No. It can be spoofed.
13. **What is SPF?**
→ Sender Policy Framework, used to help verify whether a server is authorized to send mail for a domain.
14. **What is DKIM?**
→ DomainKeys Identified Mail, which uses cryptographic signatures to help verify email integrity and domain association.
15. **What is DMARC?**
→ A domain-based email authentication policy that builds on SPF and DKIM.

L6. What information can investigators obtain from headers?

→ Routing information, timestamps, mail servers, authentication results, and sometimes IP information.

L7. What is email forensics?

→ The process of collecting and analyzing email evidence for an investigation.

L8. What is a message ID?

→ A unique identifier assigned to an email message.

L9. Can email headers be modified?

→ Some header fields can be forged or manipulated, so investigators should verify evidence using multiple indicators.

L20. What is the main purpose of this practical?

→ To analyze email headers and identify useful forensic information about an email's path and authenticity.

2. CAPTCHA Image Generation & Verification

20 Viva Questions

1. What is CAPTCHA?

→ A test designed to distinguish humans from automated programs.

2. What does CAPTCHA stand for?

→ Completely Automated Public Turing test to tell Computers and Humans Apart.

3. Why is CAPTCHA used?

→ To prevent automated bots from abusing online services.

4. What is an image CAPTCHA?

→ A CAPTCHA that displays characters or visual challenges in an image.

5. How does a CAPTCHA system work?

→ It generates a challenge, displays it to the user, and verifies the user's response.

6. **Why is random text used?**
 - To make automated prediction more difficult.
7. **Why are characters distorted?**
 - To make OCR-based automated recognition harder.
8. **What is OCR?**
 - Optical Character Recognition, which extracts text from images.
9. **What is verification?**
 - Comparing the user's answer with the expected CAPTCHA answer.
- L10. **What happens if the answer is correct?**
 - The user passes the CAPTCHA.
- L11. **What happens if the answer is incorrect?**
 - The CAPTCHA verification fails.
- L12. **Why should CAPTCHA values be randomly generated?**
 - To prevent attackers from predicting them.
- L13. **What is noise in CAPTCHA?**
 - Random visual elements added to make automated recognition harder.
- L14. **What is CAPTCHA expiration?**
 - The challenge becomes invalid after a certain period.
- L15. **What is the purpose of background distortion?**
 - To make automated image recognition more difficult.
- L16. **What is a bot?**
 - An automated program that performs tasks over a network.
- L17. **Can CAPTCHA provide complete security?**
 - No. It is one security layer and can have weaknesses.
- L18. **What is a text CAPTCHA?**
 - A CAPTCHA requiring the user to identify text or characters.

19. **Why should CAPTCHA answers not be predictable?**

→ Predictability makes automated bypass easier.

20. **What is the main objective of this practical?**

→ To generate a CAPTCHA image and verify whether the entered response is correct.

3. Wi-Fi Unauthorized Access Detection & Prevention

20 Viva Questions

1. **What is Wi-Fi?**

→ A wireless networking technology based on IEEE 802.11 standards.

2. **What is IEEE 802.11?**

→ A family of standards for wireless LAN communication.

3. **What is an Access Point?**

→ A device that provides wireless network connectivity to clients.

4. **What is SSID?**

→ Service Set Identifier; the name used to identify a Wi-Fi network.

5. **What is a Wi-Fi adapter?**

→ Hardware that allows a device to connect to a wireless network.

6. **What is unauthorized access?**

→ Access to a network without permission.

7. **How can unauthorized login attempts be detected?**

→ By monitoring authentication logs, connection attempts, and network events.

8. **What is WPA2/WPA3?**

→ Wi-Fi security protocols used to protect wireless networks.

9. **Why should strong Wi-Fi passwords be used?**

→ To reduce the risk of unauthorized access.

L0. What is MAC address filtering?

→ A method of allowing or denying devices based on their MAC addresses.

L1. Is MAC filtering alone strong security?

→ No. It should not be treated as the only security mechanism.

L2. What is an IP address?

→ An address used to identify a device/interface on a network.

L3. What is DHCP?

→ Dynamic Host Configuration Protocol, which automatically assigns network configuration such as IP addresses.

L4. What is a printer IP address used for?

→ It allows network devices to communicate with the printer.

L5. What is an authentication failure?

→ An attempt to connect that fails because the supplied credentials/security information are not accepted.

L6. How can suspicious repeated attempts be handled?

→ Through access-point security controls, monitoring, alerting, and appropriate blocking policies.

L7. Why should the experiment be performed in a controlled lab?

→ To avoid unauthorized testing of networks or devices.

L8. What is WPA3-Personal?

→ A Wi-Fi security mode using SAE-based authentication for personal networks.

L9. What is the purpose of logging connection attempts?

→ To provide evidence for detecting and investigating suspicious activity.

20. What is the main objective of this practical?

→ To understand how unauthorized Wi-Fi access attempts can be detected and prevented in a controlled IEEE 802.11 environment.

4. Recovering Permanently Deleted Files & Deleted Partitions

20 Viva Questions

1. **What is computer forensics?**
→ The process of collecting, preserving, examining, and analyzing digital evidence.
2. **What is a deleted file?**
→ A file that has been removed from the normal file-system directory structure.
3. **Does deleting a file always immediately erase its data?**
→ No. Depending on the storage technology and circumstances, data may remain recoverable until overwritten or otherwise made inaccessible.
4. **What is file recovery?**
→ The process of attempting to restore deleted or inaccessible files.
5. **What is a deleted partition?**
→ A partition whose partition-table entry or filesystem structure has been removed or damaged.
6. **What is a partition?**
→ A logical division of a storage device.
7. **What is a file system?**
→ A system used to organize and manage files on storage media.
8. **Give examples of file systems.**
→ NTFS, FAT32, exFAT, ext4.
9. **What is a forensic image?**
→ A bit-for-bit copy of digital storage used for forensic examination.
10. **Why should investigators work on a forensic copy?**
→ To preserve the original evidence.
11. **What is a write blocker?**
→ A hardware or software mechanism designed to prevent writes to evidence media during acquisition.

L2. What is data carving?

→ Recovering files based on known file structures/signatures rather than relying only on file-system metadata.

L3. What is file signature?

→ A characteristic byte pattern associated with a file type.

L4. What is a partition table?

→ Data structure describing partitions on a storage device.

L5. What is MBR?

→ Master Boot Record, an older partitioning/boot structure.

L6. What is GPT?

→ GUID Partition Table, a modern partitioning scheme.

L7. What happens if deleted data is overwritten?

→ Recovery may become impossible or significantly more difficult.

L8. Why should recovered files be verified?

→ To ensure the recovered data is complete and not corrupted.

L9. What is chain of custody?

→ Documentation showing how evidence was collected, handled, transferred, and stored.

L20. What is the main purpose of this practical?

→ To understand forensic techniques for recovering deleted files/partitions while preserving evidence integrity.

5. Log Capturing & Event Correlation

20 Viva Questions

1. What is a log?

→ A record of events generated by a system, application, or network device.

2. **What is log capturing?**
 - Collecting logs from different sources for monitoring and analysis.
3. **Why are logs important?**
 - They help detect problems, security incidents, and suspicious activities.
4. **What is an event?**
 - A recorded occurrence such as a login, error, or network connection.
5. **What is event correlation?**
 - Combining related events to identify a larger activity or possible incident.
6. **Why is event correlation necessary?**
 - Individual events may not be meaningful alone, but related events can reveal an attack or problem.
7. **Give an example of event correlation.**
 - Multiple failed logins followed by a successful login from the same source may indicate suspicious activity.
8. **What is a timestamp?**
 - The date and time associated with an event.
9. **Why is time synchronization important?**
 - It allows events from different systems to be correctly ordered and correlated.
10. **What is a log source?**
 - The system or application that generates logs.
11. **Give examples of log sources.**
 - Servers, firewalls, routers, applications, operating systems, and authentication systems.
12. **What is a SIEM?**
 - Security Information and Event Management system used to collect, analyze, and correlate security events.
13. **What is an alert?**
 - A notification generated when a rule or condition indicates potentially important

activity.

L4. What is a security incident?

→ An event or series of events that threatens security or violates security policy.

L5. What is log normalization?

→ Converting logs from different formats into a consistent structure.

L6. What is log aggregation?

→ Collecting logs from multiple sources into a central location.

L7. What is event correlation rule?

→ A condition or logic used to connect related events.

L8. What is a false positive?

→ An alert that incorrectly identifies normal activity as suspicious.

L9. What is a false negative?

→ A real suspicious event that is not detected.

20. What is the main objective of this practical?

→ To capture logs and correlate related events to identify suspicious or security-relevant activity.

6. Vulnerability Assessment — Wireshark / Snort

20 Viva Questions

1. What is vulnerability assessment?

→ The process of identifying and evaluating weaknesses in systems or networks.

2. What is Wireshark?

→ A network protocol analyzer used to capture and inspect network traffic.

3. What is Snort?

→ An open-source network intrusion detection/prevention system.

4. What is packet capture?

→ Recording network packets for analysis.

5. **What is a network packet?**
→ A unit of data transmitted over a network.
6. **What is a protocol?**
→ A set of rules used for communication between network devices.
7. **Give examples of network protocols.**
→ TCP, UDP, IP, DNS, HTTP, HTTPS.
8. **What is a packet filter/display filter?**
→ A filter used to focus analysis on selected network traffic.
9. **What is an IP address?**
→ An address identifying a network interface/device in an IP network.
- L0. **What is a TCP port?**
→ A logical endpoint used to identify a network service or application.
- L1. **What is IDS?**
→ Intrusion Detection System; detects suspicious activity and generates alerts.
- L2. **What is IPS?**
→ Intrusion Prevention System; can actively block or prevent detected malicious activity.
- L3. **What is the difference between Wireshark and Snort?**
→ Wireshark is primarily for interactive packet analysis; Snort is primarily an IDS/IPS using rules to detect suspicious traffic.
- L4. **What is a Snort rule?**
→ A rule defining traffic patterns or conditions that should trigger an alert/action.
- L5. **What is an alert?**
→ A notification that potentially suspicious traffic or activity has been detected.
- L6. **What is promiscuous mode?**
→ A network-interface mode that allows capture of packets beyond those specifically addressed to the host, subject to network hardware/configuration.

L7. **What is a vulnerability?**

→ A weakness that could potentially be exploited to affect security.

L8. **What is a false positive in intrusion detection?**

→ Normal activity incorrectly reported as malicious or suspicious.

L9. **Why is traffic analysis useful?**

→ It helps understand network behavior and investigate suspicious communications.

20. **What is the main objective of this practical?**

→ To learn how network traffic and security alerts can be monitored and analyzed using tools such as Wireshark or Snort.

7. Honeypot

20 Viva Questions

1. **What is a honeypot?**

→ A deliberately exposed system or service designed to attract and observe unauthorized activity.

2. **Why is a honeypot used?**

→ To detect, study, and analyze suspicious or malicious behavior.

3. **Is a honeypot a real production server?**

→ Usually no; it is designed primarily for security monitoring and research.

4. **What is the main purpose of a honeypot?**

→ To act as a decoy and collect information about attacks.

5. **What is a low-interaction honeypot?**

→ A honeypot that simulates a limited set of services and interactions.

6. **What is a high-interaction honeypot?**

→ A more realistic environment that allows attackers to interact with a larger set of real services.

7. **Which honeypot is generally easier to deploy?**
→ Low-interaction honeypots.
8. **Which provides more detailed attacker behavior?**
→ High-interaction honeypots.
9. **What is a honeynet?**
→ A network containing multiple honeypots.
10. **What information can a honeypot collect?**
→ Connection attempts, source information, commands, timestamps, and attacker behavior, depending on its design.
11. **What is a honeytoken?**
→ A fake piece of information or resource designed to trigger an alert when accessed.
12. **What is the difference between honeypot and firewall?**
→ A firewall controls/filters traffic, while a honeypot attracts and monitors suspicious activity.
13. **What is the difference between honeypot and IDS?**
→ IDS monitors activity across systems/networks; a honeypot is a decoy designed to attract activity for observation.
14. **What is deception in cybersecurity?**
→ Using fake systems, resources, or information to mislead attackers and detect their activity.
15. **What is an advantage of honeypots?**
→ They can generate focused security data because legitimate users normally should not interact with them.
16. **What is a disadvantage of honeypots?**
→ If poorly isolated or configured, they can introduce security risks.
17. **Why should a honeypot be isolated?**
→ To prevent an attacker from using it to reach other systems.

L8. **Can honeypots prevent all attacks?**

→ No. Their primary purpose is detection, observation, and research.

L9. **What is attacker profiling?**

→ Studying observed attacker behavior, techniques, and patterns.

20. **What is the main objective of the honeypot practical?**

→ To understand how decoy systems can attract and record suspicious activity for security analysis.

35 MOST IMPORTANT QUESTIONS FOR YOUR VIVA

If you don't have enough time to learn all **140 questions**, focus on these:

Email Forensics

1. **What is an email header?** → Metadata about an email.
2. **What is Received header?** → Shows information about mail-server handling/routing.
3. **What is email spoofing?** → Forging email identity information.
4. **What is SPF?** → Checks whether sending servers are authorized for a domain.
5. **What is DKIM?** → Cryptographic email signature/authentication mechanism.

CAPTCHA

6. **What is CAPTCHA?** → Human-vs-bot verification test.
7. **Why use random characters?** → To make automated prediction harder.
8. **What is OCR?** → Extracting text from images.
9. **Why add noise?** → To make automated recognition harder.

Wi-Fi Security

- L0. **What is IEEE 802.11?** → Wi-Fi standard family.
- L1. **What is SSID?** → Wi-Fi network name.
- L2. **What is an Access Point?** → Device providing wireless network access.
- L3. **What is unauthorized access?** → Access without permission.

L4. **Why use WPA2/WPA3?** → To secure Wi-Fi communication/authentication.

File Recovery

L5. **What is computer forensics?** → Collection and analysis of digital evidence.

L6. **What is a forensic image?** → Bit-for-bit copy of storage.

L7. **What is data carving?** → Recovering files using file structures/signatures.

L8. **What is chain of custody?** → Record of evidence handling.

L9. **What happens when deleted data is overwritten?** → Recovery may become impossible.

Logs

20. **What is a log?** → Record of system/application events.

21. **What is event correlation?** → Connecting related events to identify meaningful activity.

22. **What is SIEM?** → System for security-event collection, analysis and correlation.

23. **What is a false positive?** → Normal activity incorrectly flagged.

Wireshark/Snort

24. **What is Wireshark?** → Packet/network protocol analyzer.

25. **What is Snort?** → Network IDS/IPS.

26. **What is IDS?** → Detects suspicious activity.

27. **What is IPS?** → Detects and can block/prevent activity.

28. **What is a packet?** → Unit of network data.

Honeypot

29. **What is a honeypot?** → Decoy system that attracts and monitors suspicious activity.

30. **Why use it?** → Detection and attack analysis.

31. **Low vs high interaction?** → Limited simulation vs more realistic interaction.

32. **What is a honeynet?** → Network of honeypots.

- 33. **Why isolate a honeypot?** → To prevent compromise from spreading.
- 34. **Can a honeypot prevent all attacks?** → No.
- 35. **Main purpose?** → Detect, observe, and study unauthorized activity.

10/5/2026, 8:57:04 PM · gpt-5-6-mini



1. Write a program to Compute Similarity between two text documents. 2. Implement Page Rank Algorithm. 3. Write a program for Pre-processing of a Text Document: stop word removal. 4. Write a map-reduce program to count the number of occurrences of each alphabetic character in the given dataset. The count for each letter should be case-insensitive (i.e., include both uppercase and lower-case versions of the letter; Ignore non-alphabetic characters). 5. Write a program to implement simple web crawler. 6. Write a program to parse XML text, generate Web graph and compute topic specific page

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Absolutely. Here are **20 viva/oral questions for each of these 6 Information Retrieval practicals**, with **short, easy-to-memorize answers**.

1. Compute Similarity Between Two Text Documents

20 Viva Questions

- 1. What is document similarity?**
→ It measures how similar two documents are.
- 2. Why do we calculate document similarity?**
→ To find documents with similar content or meaning.
- 3. What is a common method for text similarity?**
→ **Cosine similarity**.
- 4. What is Cosine similarity?**
→ It measures the cosine of the angle between two document vectors.
- 5. What is the range of Cosine similarity?**
→ Usually **0 to 1** for non-negative text vectors.

6. **What does Cosine similarity = 1 mean?**
→ The documents have identical vector direction.
7. **What does Cosine similarity = 0 mean?**
→ The documents have no overlap in the represented terms.
8. **What is a document vector?**
→ A numerical representation of a document.
9. **What is TF?**
→ Term Frequency—the number of times a term occurs in a document.
10. **What is IDF?**
→ Inverse Document Frequency, which gives less weight to terms appearing in many documents.
11. **What is TF-IDF?**
→ A weighting scheme combining term frequency and inverse document frequency.
12. **Why is preprocessing needed before similarity calculation?**
→ To remove unnecessary variations and noise from the text.
13. **What is tokenization?**
→ Splitting text into individual words or tokens.
14. **What are stop words?**
→ Common words such as "the", "is", and "and" that often carry limited retrieval value.
15. **What is stemming?**
→ Reducing words to a common root-like form.
16. **What is normalization?**
→ Converting text into a consistent representation, such as lowercase.
17. **Can two documents have similar meaning but low lexical similarity?**
→ Yes, especially when they use different words or synonyms.

18. What is Euclidean distance?

→ A distance measure between vectors; it can also be used for document comparison.

19. Which similarity measure is commonly used in Information Retrieval?

→ Cosine similarity.

20. What is the main application of document similarity?

→ Search, duplicate detection, document clustering, recommendation, and plagiarism analysis.

★ **Remember:**

Text → preprocessing → vector → similarity → result

2. PageRank Algorithm

20 Viva Questions

1. What is PageRank?

→ An algorithm originally developed to rank web pages based on the link structure of the web.

2. Who developed PageRank?

→ Larry Page and Sergey Brin.

3. What is the main idea of PageRank?

→ A page is considered important when important pages link to it.

4. What type of algorithm is PageRank?

→ A link-analysis/ranking algorithm.

5. What is a web graph?

→ A graph where web pages are nodes and hyperlinks are directed edges.

6. What is a node in a web graph?

→ A web page.

7. **What is an edge?**
→ A hyperlink from one page to another.
8. **What is damping factor?**
→ A parameter representing the probability that a user follows links rather than randomly jumping to another page.
9. **What is the commonly used damping factor?**
→ **0.85.**
- L0. **Why is damping factor used?**
→ To model random jumps and help ensure stable ranking.
- L1. **What is the PageRank value?**
→ A numerical score representing the importance of a page in the link graph.
- L2. **Does having many incoming links always guarantee the highest PageRank?**
→ No. The importance of linking pages and the overall link structure also matter.
- L3. **What is an incoming link?**
→ A link pointing to a page.
- L4. **What is an outgoing link?**
→ A link from a page to another page.
- L5. **What is an iterative algorithm?**
→ An algorithm that repeatedly updates values until a stopping condition is reached.
- L6. **Why is PageRank calculated iteratively?**
→ Because each page's rank depends on the ranks of other pages.
- L7. **What is convergence?**
→ When PageRank values stop changing significantly between iterations.
- L8. **What happens if a page has no outgoing links?**
→ It is a dangling node and needs special handling in PageRank calculations.

19. **Where is PageRank used?**

→ Web search and link analysis.

20. **What is the main limitation of basic PageRank?**

→ It uses link structure and does not directly represent the actual textual relevance of a query.

3. Stop Word Removal

20 Viva Questions

1. **What is stop word removal?**

→ Removing common words that may contribute little to information retrieval.

2. **Give examples of stop words.**

→ "the", "is", "a", "an", "and", "of".

3. **Why remove stop words?**

→ To reduce unnecessary terms and improve processing efficiency.

4. **What is preprocessing?**

→ Preparing raw text for further analysis.

5. **What is tokenization?**

→ Breaking text into individual tokens or words.

6. **Why is tokenization performed?**

→ To process individual words separately.

7. **What is a stop word list?**

→ A predefined collection of common words that may be removed.

8. **Are stop words always useless?**

→ No. Their importance depends on the application and context.

9. **Can removing stop words change the meaning of a sentence?**

→ Yes, in some cases.

L0. What is case folding?

→ Converting text to a common case, usually lowercase.

L1. What is stemming?

→ Reducing words to a common stem.

L2. What is lemmatization?

→ Converting words to their dictionary/base form.

L3. Is stemming the same as stop word removal?

→ No. Stemming changes word forms, while stop word removal removes selected words.

L4. What is the benefit of preprocessing?

→ It reduces noise and can improve efficiency.

L5. What happens if we don't remove stop words?

→ The vocabulary can contain many frequent, less informative terms.

L6. Can stop word lists differ between languages?

→ Yes.

L7. Can stop word removal reduce storage requirements?

→ Yes, because fewer terms need to be stored/indexed.

L8. Where is stop word removal used?

→ Search engines, text classification, NLP, and information retrieval.

L9. Should every common word be removed?

→ No. The decision should depend on the task.

L20. What is the main objective of this practical?

→ To preprocess a text document by identifying and removing selected stop words.

4. MapReduce — Alphabet Character Frequency

20 Viva Questions

1. **What is MapReduce?**
→ A programming model for processing large datasets using map and reduce operations.
2. **What does the Map phase do?**
→ It processes input data and generates intermediate key-value pairs.
3. **What does the Reduce phase do?**
→ It combines values associated with the same key.
4. **What is the key in this practical?**
→ An alphabetic character.
5. **What is the value?**
→ The count associated with that character.
6. **Why is the count case-insensitive?**
→ Uppercase and lowercase versions of the same letter should have the same total count.
7. **What happens to non-alphabetic characters?**
→ They are ignored.
8. **What is a mapper?**
→ The component that transforms input records into intermediate key-value pairs.
9. **What is a reducer?**
→ The component that aggregates values for each key.
10. **What is the intermediate key-value pair?**
→ For example, (a, 1) for each occurrence of the letter a.
11. **What does shuffle and sort do?**
→ It groups intermediate values by key before reduction.
12. **Why convert characters to lowercase?**
→ To make uppercase and lowercase counts case-insensitive.

L3. Why use MapReduce?

→ It can distribute large-scale data processing across multiple machines.

L4. What is a dataset?

→ A collection of input data to be processed.

L5. What is Hadoop?

→ An open-source framework commonly associated with distributed storage and MapReduce processing.

L6. What is HDFS?

→ Hadoop Distributed File System.

L7. Is MapReduce sequential or parallel?

→ It is designed for distributed and parallel processing.

L8. What is the final output?

→ The frequency/count of each alphabetic character.

L9. Why are non-alphabetic characters ignored?

→ Because the problem specifically asks for alphabetic character frequencies.

20. What is the main advantage of MapReduce?

→ It enables scalable processing of large datasets.

 Example

Input:

Apple A!

After case normalization:

apple a

The output counts would include:

a → 2

5. Simple Web Crawler

20 Viva Questions

1. **What is a web crawler?**
→ A program that automatically visits web pages and collects information.
2. **What is another name for a web crawler?**
→ Spider or bot.
3. **What is a seed URL?**
→ The initial URL from which crawling begins.
4. **What does a crawler collect?**
→ Web pages, links, and selected information from pages.
5. **What is a URL?**
→ Uniform Resource Locator; an address identifying a resource on the web.
6. **What is a queue in a crawler?**
→ A data structure used to store URLs waiting to be visited.
7. **Why is a queue useful?**
→ It can support breadth-first crawling.
8. **What is BFS?**
→ Breadth-First Search, which processes items level by level.
9. **What is the purpose of a visited set?**
→ To prevent repeatedly crawling the same URL.
10. **What happens after visiting a page?**
→ The crawler can extract links and add new URLs to the queue.
11. **What is link extraction?**
→ Finding hyperlinks in an HTML page.
12. **Why should a crawler have a stopping condition?**
→ To prevent unlimited crawling.
13. **What is robots.txt?**
→ A file that communicates crawling preferences/rules to web crawlers.

L4. What is HTTP?

→ Hypertext Transfer Protocol used for web communication.

L5. What is HTML?

→ HyperText Markup Language used to structure web pages.

L6. What is a web crawler library in Python?

→ A library that helps fetch and/or parse web content, such as Requests or BeautifulSoup for fetching/parsing tasks.

L7. What is BeautifulSoup used for?

→ Parsing HTML/XML and extracting information from structured markup.

L8. What is the difference between a browser and a crawler?

→ A browser is designed for interactive human use; a crawler automatically retrieves and processes pages.

L9. What problem can occur if visited URLs aren't tracked?

→ The crawler may repeatedly visit the same pages and enter loops.

20. What is the main objective of this practical?

→ To build a basic program that starts from a seed URL, visits pages, and extracts links.

6. XML Parsing, Web Graph & Topic-Specific PageRank

20 Viva Questions

1. What is XML?

→ Extensible Markup Language used to represent structured data.

2. What is XML parsing?

→ Reading and extracting structured information from XML data.

3. What is an XML element?

→ A logical component represented using opening and closing tags.

4. **What is an XML attribute?**
→ Additional information associated with an XML element.
5. **What is a web graph?**
→ A graph representing web pages and links between them.
6. **What is a node in a web graph?**
→ A web page or document.
7. **What is an edge?**
→ A link from one page/document to another.
8. **What is PageRank?**
→ A link-based algorithm for assigning importance scores to pages.
9. **What is topic-specific PageRank?**
→ A variation of PageRank that gives greater importance to pages related to a particular topic.
10. **How is topic-specific PageRank different from normal PageRank?**
→ It biases the ranking toward a selected topic instead of treating all pages equally in the teleportation process.
11. **What is a topic?**
→ A subject or category of interest used to focus the ranking.
12. **What is a teleportation vector?**
→ A probability distribution indicating where the random-surfer model can jump.
13. **Why is the teleportation vector important in topic-specific PageRank?**
→ It biases the ranking toward topic-relevant pages.
14. **What is damping factor?**
→ A parameter controlling link-following versus teleportation.
15. **What is the common damping factor?**
→ **0.85.**

- L6. **Why do we parse XML before generating the graph?**
→ To extract pages and their link relationships from structured data.
- L7. **What is a directed graph?**
→ A graph in which edges have a direction.
- L8. **Why is a web graph directed?**
→ Because a hyperlink points from one page to another.
- L9. **What is convergence in PageRank?**
→ When successive PageRank iterations become sufficiently stable.
20. **What is the main objective of this practical?**
→ To parse XML data, construct a web graph, and calculate rankings biased toward a specific topic.



MOST IMPORTANT VIVA QUESTIONS TO MEMORIZE

If your examiner asks only a few questions, **focus on these**:

Practical	Most important points
Document Similarity	Cosine similarity, TF-IDF, tokenization, vectors
PageRank	Nodes, edges, damping factor, incoming links, iterative calculation
Stop Words	Definition, examples, purpose, tokenization, stemming
MapReduce	Mapper, reducer, key-value pair, shuffle/sort, case-insensitive count
Web Crawler	Seed URL, queue, visited set, link extraction, robots.txt
Topic PageRank	XML parsing, web graph, topic, teleportation vector, damping factor



10 one-line answers you should definitely know



1. **Cosine similarity** → Measures similarity between document vectors.
2. **TF-IDF** → Gives higher weight to terms important to a document but less common across documents.
3. **PageRank** → Ranks pages using the web's link structure.
4. **Damping factor** → Controls link following versus random teleportation.
5. **Stop word** → A common word that may be removed during preprocessing.
6. **Mapper** → Produces intermediate key-value pairs.
7. **Reducer** → Aggregates values for the same key.
8. **Seed URL** → Starting URL of a crawler.
9. **Visited set** → Prevents repeated crawling of the same URL.
10. **Topic-specific PageRank** → PageRank biased toward a selected topic.

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